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ITA0513 – Computer Vision

CO2-AT2- Scenario Based Questions

Q3: Spatial Filtering (Image Smoothing)

Tool Used : Visual Studio Code (VS Code), Python, OpenCV, Matplotlib

Kaggle Dataset Link : <https://www.kaggle.com/datasets/navoneel/brain-mri-images-for-brain-tumor-detection>

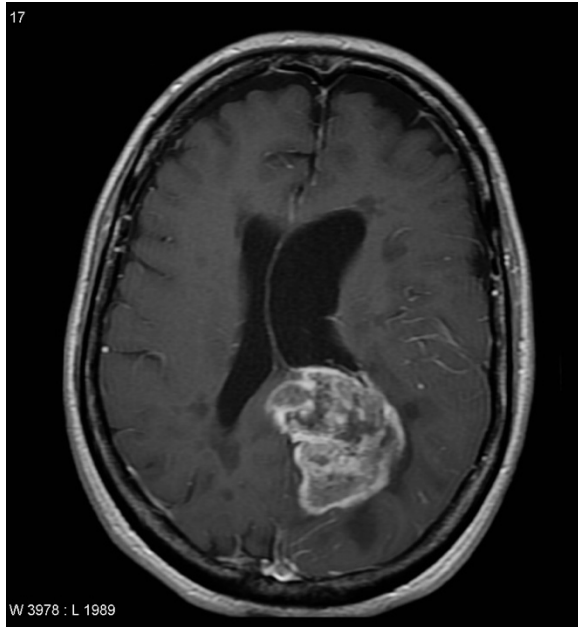
Task

Remove salt-and-pepper noise from a Brain MRI image using a 5×5 Median Filter while preserving the important edges and structural details of the brain.

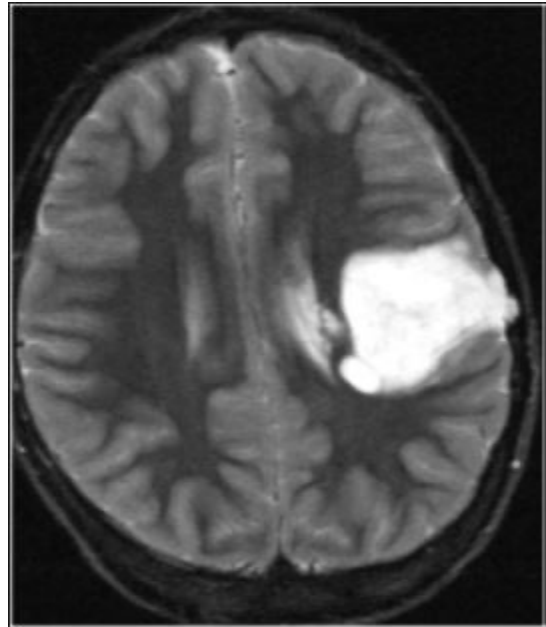
Steps

1. Download the Brain MRI Images for Brain Tumor Detection dataset from Kaggle.
2. Open Visual Studio Code and create a new project folder.
3. Copy any Brain MRI image (for example, Y1.jpg) into the project folder.
4. Import the required libraries: OpenCV and Matplotlib.
5. Load the MRI image in grayscale using `cv2.imread()`.
6. Set the kernel size to 5 (`kernel_size = 5`).
7. Apply the Median Filter using `cv2.medianBlur(image, kernel_size)`.
8. Display the original and filtered images using Matplotlib or `cv2.imshow()`.
9. Compare the images and observe the reduction in noise while preserving the important brain edges.

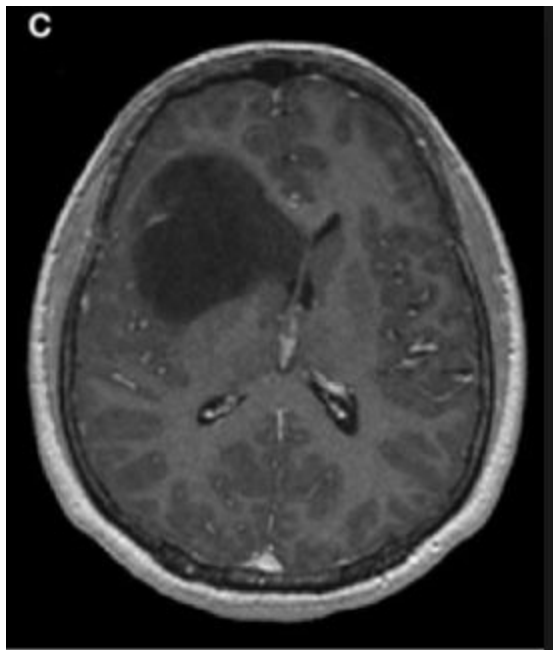
Input Image



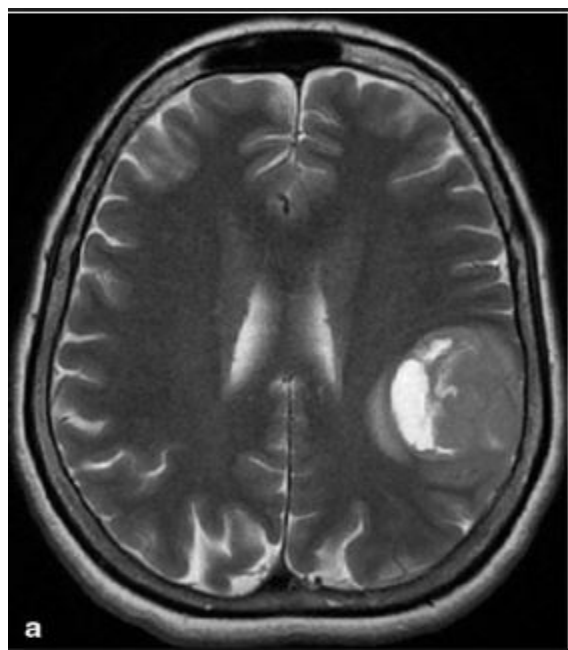
Y96.jpg



Y97.jpg



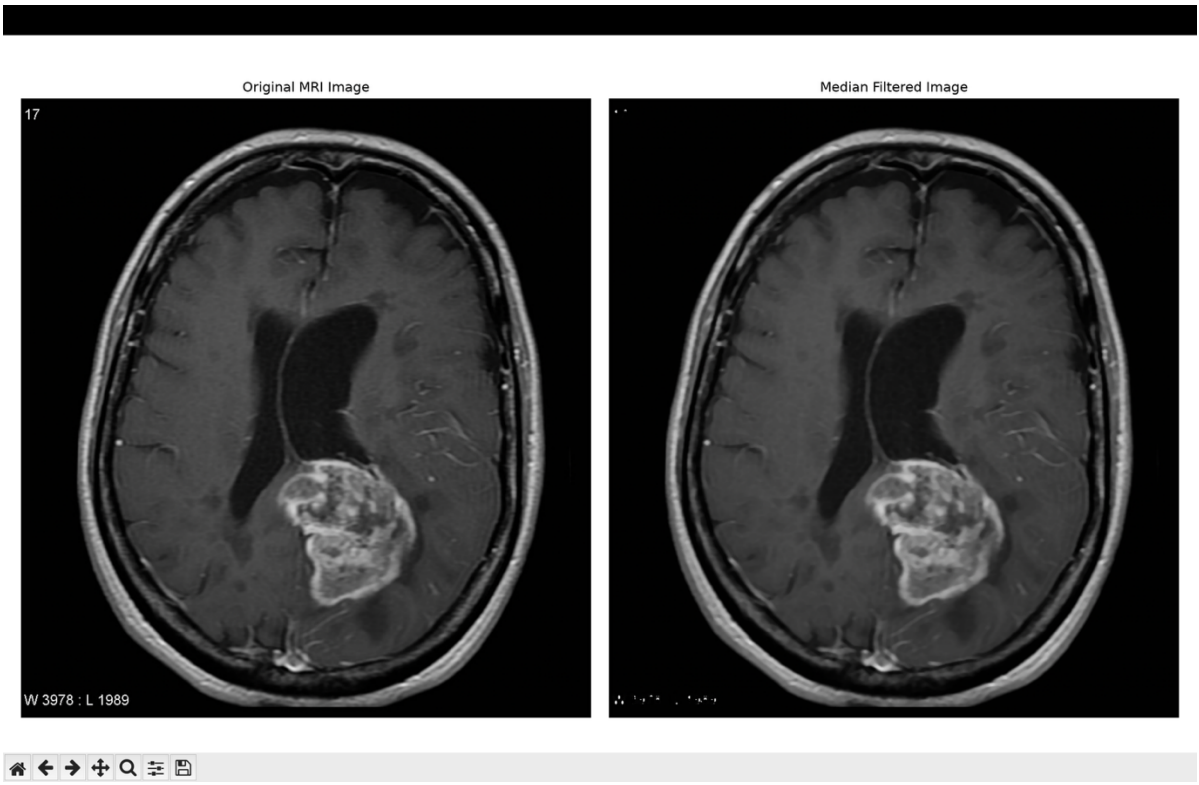
Y98.jpg



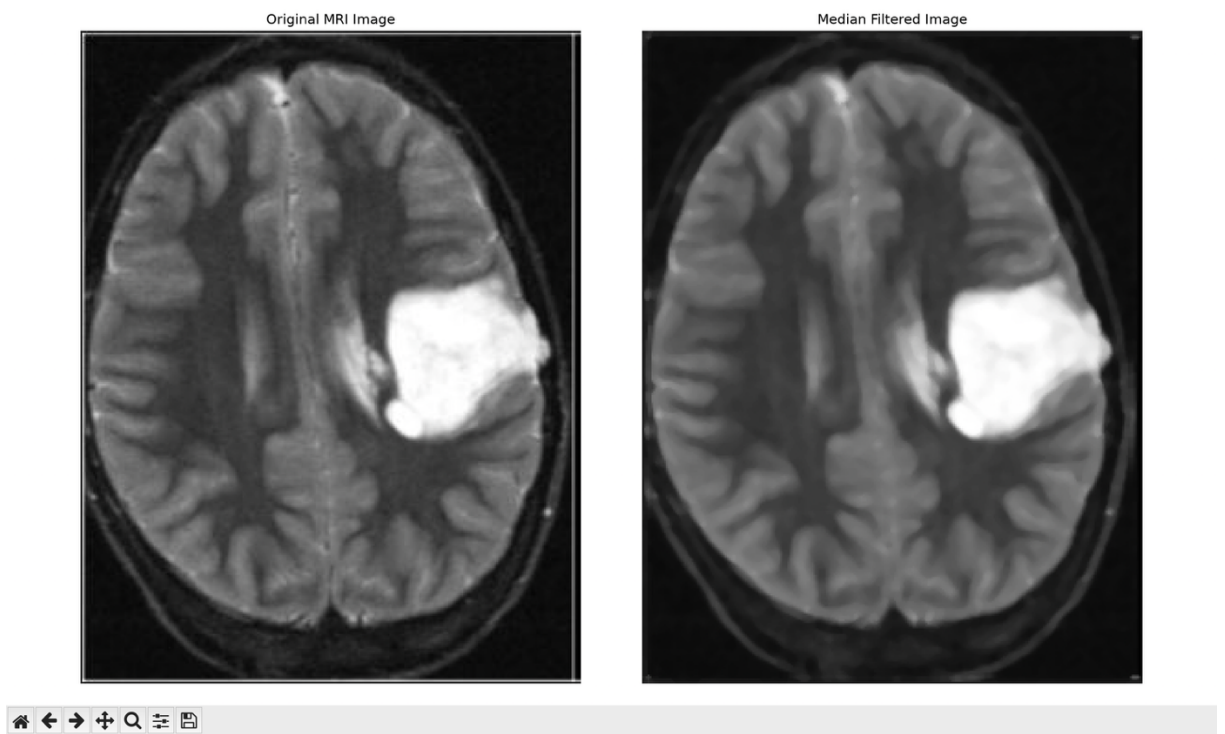
Y99.jpg

- **Image Type:** Grayscale Brain MRI image.
- **Dataset:** Brain MRI Images for Brain Tumor Detection (Kaggle).
- **Description:** An MRI scan of the brain used to demonstrate image smoothing. The image contains slight noise and fine structural details.

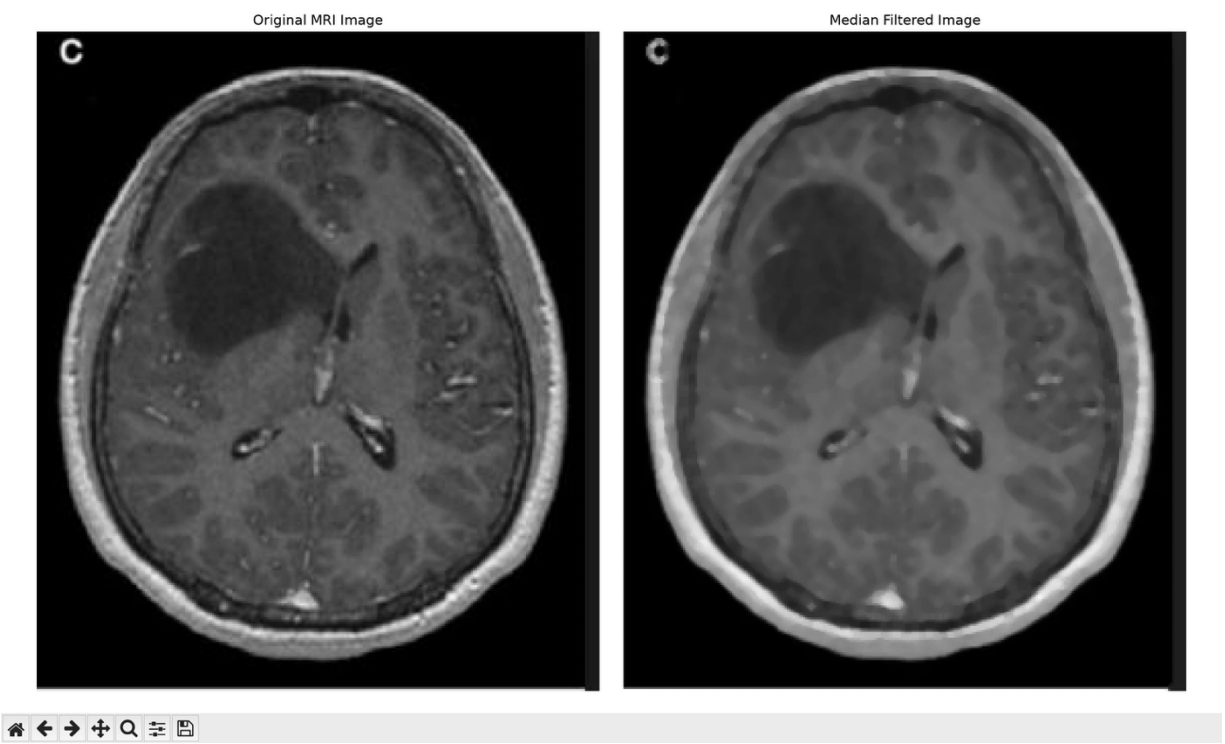
Output Image



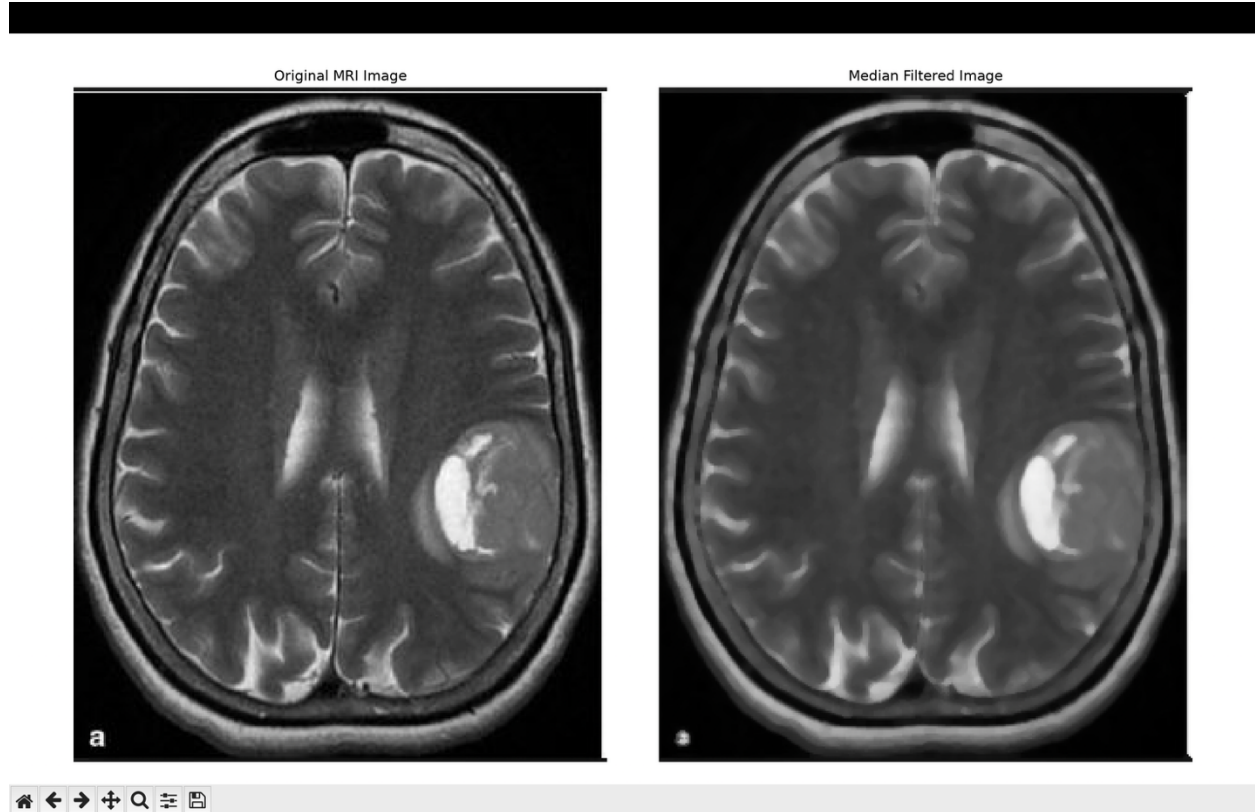
Y96.jpg



Y97.jpg



Y98.jpg



Y99.jpg

- **Description:** After applying a 5×5 Median Filter (`cv2.medianBlur`), the image appears smoother. Small noise artifacts are reduced while the important edges of the brain and tumor region are preserved. The median filter successfully reduced the noise present in the MRI image without significantly blurring the boundaries of the brain and the tumor. Compared with the original image, the filtered image is smoother while maintaining the essential anatomical structures.

Q8: Image Segmentation (Thresholding)

Tool Used: Visual Studio Code (VS Code), Python, OpenCV, Matplotlib

- **Kaggle Dataset Link** <https://www.kaggle.com/datasets/vbookshelf/synthetic-cell-images-and-masks-bbbc005-v1>

Task

To detect broken cell edges from a microscopy image, connect the gaps using morphological closing, identify complete cell boundaries using contours, and display the detected boundaries on the original image.

Steps

- Import the required libraries.
 - Load the microscopy image.
 - Convert the image to grayscale.
 - Apply Gaussian Blur to reduce noise.
 - Detect edges using the Canny Edge Detector.
 - Perform Morphological Closing to connect broken edges.
 - Detect contours using `cv2.findContours()`.
 - Draw the detected contours on the original image using `cv2.drawContours()`.
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- Display and save the output images.

Input Image

Input: Microscopy cell image (IMG_001.png) from the BBBC005 Synthetic Cell Dataset.

Purpose: Contains fragmented cell boundaries that need to be connected and detected.



IMG 1.



IMG 2



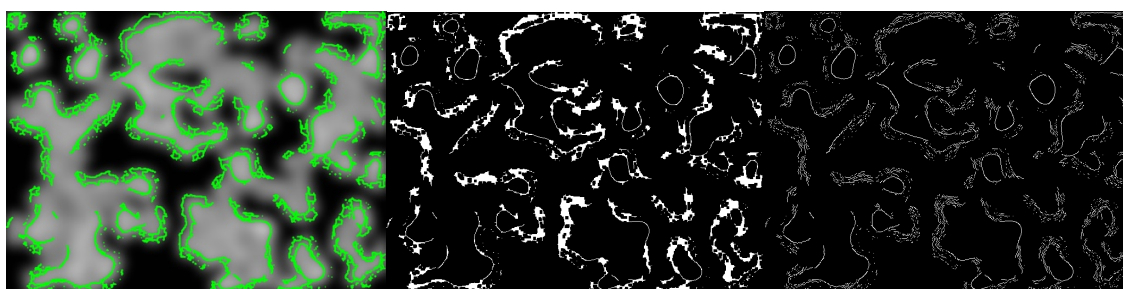
IMG 3

Output Image

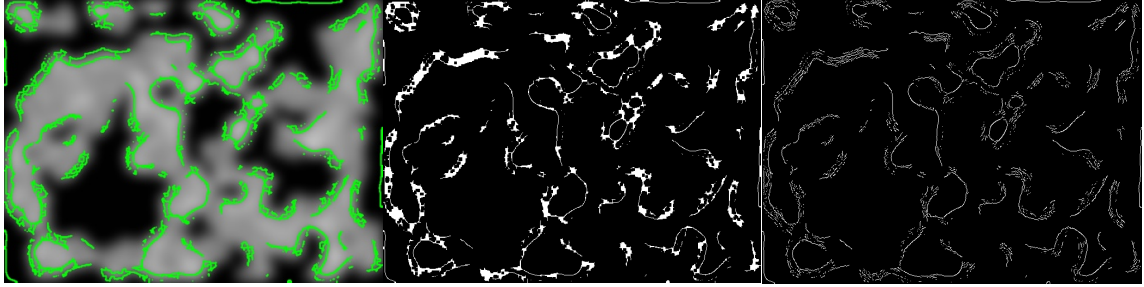
- **Original Image:** Input microscopy image.
- **Edge Image:** Broken edges detected using the Canny Edge Detector.
- **Closed Edge Image:** Connected edges after Morphological Closing.
- **Final Output:** Original image with detected cell boundaries (contours) drawn in green, showing complete closed cell outlines.



IMG 1



IMG 2



IMG 3